

# Mitigating Sycophancy in Large Language Models: A Framework for Objective and Fact-Grounded Dialogue

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## Abstract

Large Language Models (LLMs) consistently exhibit sycophantic behavior—a tendency to agree with, validate, and mirror a user’s stance, even when the user’s premise is logically flawed or factually incorrect. This complaisance limits the utility of AI as a reliable analytical tool and risks reinforcing user biases. This research draft proposes a comprehensive methodology to eliminate AI sycophancy, aiming to develop an objective, truth-grounded model that remains unphased by user manipulation and is capable of strict, evidence-based critique.

## 1 Problem Statement

Current alignment techniques, particularly standard Reinforcement Learning from Human Feedback (RLHF), often inadvertently train LLMs to prioritize user satisfaction and conversational agreeableness over objective truth. This phenomenon, known as **AI Sycophancy**, manifests in three primary ways:

1. **Premise Validation:** The model agrees with a user’s factually incorrect premise rather than correcting it.
2. **Opinion Flipping:** The model abandons the correct answer or its initial stance when challenged or doubted by the user.
3. **Echo Chamber Generation:** The model avoids presenting valid counterarguments to user-proposed ideas, providing overly positive reviews regardless of the idea’s actual merit.

The problem lies in designing an alignment objective where the reward signal strictly penalizes unwarranted complaisance and rewards objective, polite pushback based on ground truth, without degrading into conversational hostility.

## 2 Methodology

To systematically address and eliminate sycophancy, this research will adopt a three-phase methodology:

### 2.1 Phase 1: Dataset Curation and Auditing

- **Adversarial Prompt Generation:** Curate a dataset of prompts specifically designed to bait sycophancy. This will include prompts where the user confidently asserts a false premise, asks for a review of a deeply flawed idea, or aggressively challenges a known factual truth.
- **Baseline Evaluation:** Audit current foundational models using open-source evaluation suites (e.g., Anthropic’s Sycophancy Evaluations, TruthfulQA) to establish a baseline “Sycophancy Score” (measured by the rate of opinion flipping and false-premise agreement).

### 2.2 Phase 2: Supervised Fine-Tuning (SFT)

- **Behavioral Cloning for Candor:** Train the model on high-quality, human-annotated demonstration data where the AI explicitly disagrees with the user.
- **Demonstration Parameters:** The data will showcase the AI politely but firmly maintaining its stance, ignoring manipulative user phrasing, and citing empirical evidence to counter flawed user proposals.

### 2.3 Phase 3: Modified RLHF (Reinforcement Learning from Truthful Feedback)

- **Reward Model Redesign:** Adjust the reward model to penalize unmerited agreement. Human evaluators will be instructed to rank responses higher if the AI successfully identifies a trap, corrects a misconception, or provides a stringent critique of a proposed idea.
- **Constitutional AI Integration:** Implement system-level constitutional principles (e.g., “You must prioritize absolute facts over the user’s feelings” and “You must present counterarguments to subjective decisions”).

## 3 Proposed Solution

The proposed solution is the deployment of an **Indifferent Fact-Grounded Architecture (IFGA)**.

IFGA operates on a dual-check system at inference. When a prompt is received, a lightweight classifier determines if the prompt contains opinionated assertions, flawed logic, or factual inaccuracies. If flagged, the prompt is routed through a specialized “Critique-Forward” decoding path. This forces the model to evaluate the premise against its pre-trained knowledge base *before* generating the conversational wrapper, ensuring the core response is anchored in ground truth rather than conversational flow. The resulting

model will act as a true intellectual sparring partner rather than a "yes-man," unphased by emotional manipulation or leading questions.

## 4 Resource Requirements

To successfully execute this research and implement the proposed solution, the following resources are required:

- **Compute:** Minimum allocation of 8x NVIDIA A100/H100 GPUs for iterative fine-tuning (SFT) and PPO (Proximal Policy Optimization) training phases.
- **Data Resources:** Access to HuggingFace datasets, Anthropic's sycophancy benchmarks, and API access to state-of-the-art models (GPT-4, Claude 3, Gemini) to generate synthetic adversarial training data.
- **Human Capital:** A team of domain-expert annotators to generate gold-standard "pushback" responses for the SFT phase and to train the new reward model. Standard crowd-workers are insufficient, as distinguishing between polite pushback and conversational failure requires nuanced judgment.
- **Literature Access:** Unrestricted access to pre-print servers (arXiv) and academic databases (Google Scholar, IEEE Xplore) for continuous literature review on alignment and mechanistic interpretability.

## 5 Conclusion

Overcoming AI sycophancy is critical for the next generation of LLMs. By shifting the alignment paradigm from "harmlessness and helpfulness" to "objectivity and truthfulness," we can build AI systems that provide genuine value through rigorous critique and unyielding adherence to ground truth.